

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

Student's name: **Thakur, Amey Mahendra**

Student ID number: **110107589**

Final Exam Question on **GA**:

Q2a. Solution:

After reproduction & Before reproduction

Generation 1 - Reproduction Process:

i	Initial Population	Value x_i	Fitness $f(x_i)$	$\frac{f(x_i)}{\sum_{i=1}^6 f(x_i)}$	Roulette Wheel Copies	New Parents	Fitness
1	001.0000	1.00	0.1600	0.0077	1	1.00	0.1600
2	010.0000	2.00	0.4442	0.0215	1	2.00	0.4442
3	011.0100	3.25	15.7480	0.7609	2	3.25	15.7480
4	100.0000	4.00	3.9841	0.1925	1	4.00	3.9841
5	101.1000	5.50	0.2499	0.0121	1	5.50	0.2499
6	110.1000	6.50	0.1111	0.0054	0	3.25	15.7480

Fitness Sum = 35.3342; *Fitness Average* = **5.8890**

% Reproduction Process:

values = [1.00, 2.00, 3.65, 3.60, 5.50, 6.50];

```

k = 1:length(values);

ini_pop = [];
for i = 1:length(values)
    d2b = string(fix(rem(values(i)*pow2(-(3-1):4),2)));
    dec = join(d2b(1:3), "");
    frac = join(d2b(4:7), "");
    ini_pop(end+1) = join([dec, frac], ".");
end

fitness = [];
for i = 1:length(values)
    fitness(end+1) = 1 / (values(i)^2 - 7*values(i) + 12.25 + 0.001);
end

rwc = [1 1 2 1 1 0];

new_ps = values;
for i = 1:length(rwc)
    if rwc(i) == 0
        new_ps(i) = values(find(rwc==2));
    end
end

[k' ini_pop' values' fitness' (fitness ./ sum(fitness))' rwc' new_ps']
T = array2table([k' ini_pop' values' fitness' (fitness ./ sum(fitness))' rwc'
new_ps'], ...
    "VariableNames", {'k' 'Ini Pop' 'Values (xi)' 'Fitness f(xi)'
'f(xi)/sum(f(xi))' 'RWCs' 'New Parents'})

new_fitness = [];
for i = 1:length(new_ps)
    new_fitness(end+1) = 1 / (new_ps(i)^2 - 7*new_ps(i) + 12.25 + 0.001);
end
fitness_sum = sum(new_fitness)
fitness_avg = sum(new_fitness) / length(new_fitness)

```

Decimal to Binary

```

function binaryFraction = decimalToBinaryFraction(decimalFraction)
    integerPart = floor(decimalFraction);
    fractionalPart = decimalFraction - integerPart;
    binaryInteger = dec2bin(integerPart);
    binaryFraction = '';
    for i = 1:16
        fractionalPart = fractionalPart * 2;
        if fractionalPart >= 1
            binaryFraction = [binaryFraction '1'];
            fractionalPart = fractionalPart - 1;
        else
            binaryFraction = [binaryFraction '0'];
        end
        if fractionalPart == 0
            break;
        end
    end
    binaryFraction = [binaryInteger '.' binaryFraction];
end

% Example usage:
x = 3.75;
y = decimalToBinaryFraction(x);
disp(['Decimal of ', num2str(x), ' is in binary ', y]);

```

Binary to Decimal

```

function decimalFraction = binaryToDecimalFraction(binaryFraction)
    parts = strsplit(binaryFraction, '.');
    decimalInteger = bin2dec(parts{1});
    decimalFractional = 0;
    for i = 1:length(parts{2})
        if parts{2}(i) == '1'
            decimalFractional = decimalFractional + 2^(-i);
        end
    end
    decimalFraction = decimalInteger + decimalFractional;
end

% Example usage:
y = '11.11';
x = binaryToDecimalFraction(y);
disp(['Binary ', y, ' is in decimal ', num2str(x)]);

```